Offline workbench: public records and analytical fixtures

What can be run now

This workbench is executable evidence for **public-record and analytical checks**. It needs no cluster credentials, model weights, corpus ingestion or GPU. It does not train a model, reproduce a paper's entire study or qualify a trainer.

```
python -m pip install -r scripts/requirements-kb.txt
python scripts/kb_workbench.py --all
python scripts/kb_workbench.py --all --check
```

The first command sequence regenerates seven receipts under `knowledge/receipts/`. `--check` recomputes results and checks them against the committed receipts without rewriting them. Inputs carry SHA-256 hashes and each receipt identifies the exact workbench-script digest. A code/input change therefore requires a deliberate new receipt, even if its visible headline result remains the same.

The seven recipes and their boundaries

Recipe	Actual input	Local check	What it does not establish
`step-law-release`	Verbatim dense/MoE CSVs from the pinned official archive	1911/708rows, declared groups, best observed grid points, sequence→token batch conversion	Continuous optima, complete original logs, causal recipe equivalence
`hf-mixture-records`	Pinned Harrier Parquet release	116rows;72fresh/44resumed; nonnegative phase-simplex weights	Possession of training text or independence of all rows
`phase-exposure`	Pinned hero mixture JSON	200cells, three phases, cumulative exposure/cap arithmetic	A measured corpus scan, phase deployment or an actual training run
`identifiability`	Explicit synthetic log-linear designs	Rank deficiency of a fixed-TPP ray versus a crossed grid	Empirical exponents or the correct nonlinear loss model
`optimizer-timescale`	Gradient-free constant-LR recurrence	Exact versus approximate batch/decay compensation	A trained AdamW optimum or transfer to MuonH
`evaluation-denominators`	Synthetic loss/token/byte sums and context budget	Macro versus pooled NLL; prompt-plus-output fit	Model capability scores or real data coverage
`compute-ledger`	Fully declared illustrative MoE dimensions and budget assumptions	Stored matrices, executed work, attention term and GPU-day ledger	Measured throughput, qualified recurrence or an optimal hero shape

The source card lists the actual files. The input provenance manifest identifies the upstream archive member or pinned repository path. The original archives remain in `references/`.

Inspect one empirical-record example

```
python scripts/kb_workbench.py step-law-release
python scripts/kb.py claim claim-steplaw-release-rows
```

The dense1.0737B/20B-token group yields a best observed smooth loss2.2254960 at LR0.001381 and sequence batch256. The2048-token context makes that524288tokens per batch. This row is descriptive finite-grid evidence under the released recipe. The checker groups by exposed shape fields; missing recipe metadata still limits causal interpretation.

Use this replay to learn the data contract before reproducing a training cell. A genuine training reproduction additionally needs the original data/tokenization, implementation, initialization/optimizer/schedule, numerical precision, hardware/runtime and evaluation definitions. A fitting repository is not automatically that complete training stack.

Inspect one Marin record and its target exposure

```
python scripts/kb_workbench.py hf-mixture-records
python scripts/kb_workbench.py phase-exposure
```

The legacy export's116observations and the newer transfer registry's804/56/115/3tables describe different scopes. Do not add them as independent runs. An18.75Tfull-scale target field simulates desired exposure; the physical proxy did not necessarily consume18.75Ttokens. Shared-prefix cooldown runs reuse a history, which matters for cost and uncertainty.

The later hero JSON contains **three** phases, whereas some legacy search records contain one or two. The workbench checks that exact later configuration instead of silently flattening all generations into "the Marin mixture".

Reading a receipt correctly

A receipt's `kind` can be `public_record_replay`, `public_configuration_check` or `analytical_fixture`. Its `result` records measurements or calculations from that specific input. The `execution` explicitly says local CPU only. A page's claim links identify which parts of the result it relies on.

The original source's reported training results remain reported. For example, checking the HF table's file hash and row count does not independently reproduce the loss stored in a row. Similarly, recomputing a candidate's exposure fractions does not prove the referenced cloud store exists or its corpus was deduplicated as intended.

From these checks to real experiments

1. Rebuild a small real-data subset and save source/normalized/token/attribute/store lineage.
2. Qualify trainer correctness, optimizer-group semantics, checkpoint/export parity and evaluation on actual hardware.
3. Measure representative end-to-end throughput and memory, including routing/drop behavior and resume continuity.
4. Run matched controls and a predeclared larger holdout, saving complete per-run records.
5. Add a training reproduction receipt only when inputs, environment, outputs and agreement tolerances are reviewable.

No current CPU receipt satisfies step5. The validator rejects `training_reproduced` claims under the current policy rather than accepting a label without a training-artifact contract.